Border Detection of Document Images Scanned From Large Books

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***Abstract* –Document images usually suffer from some unwanted transformations like skew and warping. When dealing with large books, another problem is also introduced; when we capture a page of a large book using digital camera or scanner, some extra margins appears. The resulting document is often framed by a dark border and noisy text regions from neighboring page. In this paper, we introduce a novel technique for enhancing the document images by automatically detecting the document borders and cutting out noisy area. Our methodology is based on projecting profiles combined with an edge detection process. Experimental results on several document images, mainly historical with a small slope, indicate the effectiveness of the proposed technique.**

Keywords -Border Detection, Edge Detection, OCR Preprocessing, Projection Profile

I. INTRODUCTION

Nowadays we are living in a world that is almost entirely digital. Paper documents are an exception because we prefer to browse the papers instead of watching through the monitor and boring our eyes. However, like it or not, we want to have a digital copy of our documents because of its advantages over papers. For example digital copies can be saved more easily, they can be converted to digital texts using OCR systems and then it can be searched, edited and published online.

The first step in every OCR system is preprocessing documents to produce a clean image of the document. Document images captured by a digital camera or a scanner often framed by a noisy border or text regions from neighboring pages. Detecting and removing these unwanted areas is critical to achieve better OCR results.

There are few techniques for page border detection. Fletcher et al. [1] described a bottom-up approach algorithm based on typesetting knowledge. This algorithm is robust, but it is very slow. Fan et al. [2] proposed a method for scanned documents by reducing the resolution of the document and detection of marginal noise. Avila and Lins [3] proposed invading and non-invading border algorithms based on "flood-fill" algorithms. Avilla et al. [4] proposed algorithm which works based on "flood-fill" algorithm and connected component labeling and region adjacency graphs for removing dark borders. Le and Thoma [5] proposed another method for border removal which is based on classification of black border, non-textual and textual rows and columns, location of border objects, and an analysis of projection profiles and crossing counts of textual squares.

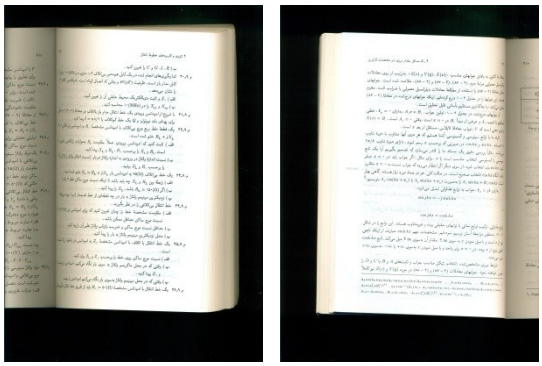
Most of the previous works detect borders and do not deal with the noisy text regions from neighboring pages. Stamatopoulos et al. [6] applied projection profile and connected component labeling for border detection and cross correlation for detecting noisy extra text regions. This method uses local binarization, which requires a lot of processing time. And finally Shafait et al. [7] tried to find the actual page contents area, ignoring marginal noise along the page border. They rely on geometric matching algorithms to find the optimal page border of structured documents by exploiting their text alignment property.

II. DATASET

To evaluate the proposed method we required a dataset of Persian documents. Since there were no such dataset, we attempted to scan some books and produce an appropriate dataset.

We scanned 159 pages of several books at 300 dpi in color format with an HP Scanjet G3110 scanner. Two samples of the dataset are shown in Fig. 1.

As shown in Fig. 1 some documents located at the left and some at the right pages. We refer to these as type 1 and type 2 documents.



(a) (b)

Fig. 1. Sample images with extra border and texts. (a) Type 1. (b) Type 2.

There were 96 pages of type 1 and 63 pages of type 2. Documents collected from these books: “Digital Logic Circuit Analysis and Design”, “Field and Wave Electromagnetics”, “Communication Systems-Carlson”, “Linear Control - Aryaz Baradarani” and some other books in Farsi language.

III. PROPOSED METHOD

The proposed algorithm for detection and elimination of dark borders and noisy text region is illustrated in Fig. 3. It is based on edge detection and projection profile and is described in the following sub-sections.

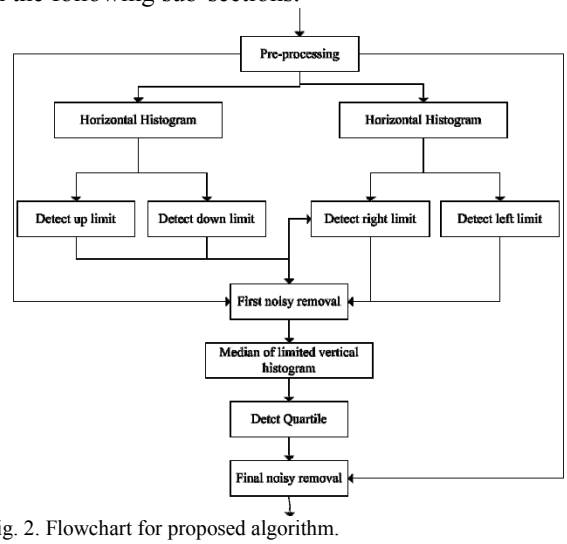


Fig. 2. Flowchart for proposed algorithm.

A. Pre-processing

The aim of the pre-processing stage is to eliminate noisy areas, smooth the background texture and enhance the contrast between background and text areas. The following operations are conducted: At first, an adaptive wiener filter [8] is applied on the grayscale source image I_s according to the Eq. 1:

$$\mathrm{I}(\mathrm{x},\mathrm{y}) = \mu + ((\sigma^2 - v^2)(I_s(x,y) - \mu))/\sigma^2 \qquad (1)$$

Where μ is the local mean, σ^2 the variance at 3×3 neighborhood around each pixel and v^2 is the average of all estimated variances for each pixel in the neighborhood.

B. Edge detection

In this stage, horizontal and vertical edges detected based on Prewitt operator [9]. As shown in Fig. 3 Weiner filter reduces noise especially in the border area.

C. Horizontal Histogram

Calculate horizontal histogram, which is sum of black pixels in each row.

$$H_h(y) = \sum_{x=1}^{I_x} P(x,y)\ ,\ where\ 1 \le y \le I_y \qquad (2)$$

Which I_x is number of columns in the document image, $P(x,y)$ is the result of applying Prewitt operator to the filtered grayscale image I.

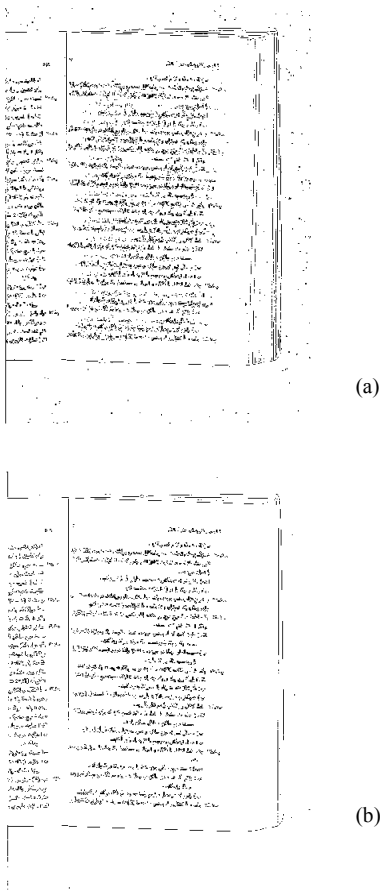


Fig. 3. Result of edge detection algorithm on sample image of Fig. 1. (a) Before wiener filter. (b) After wiener filter.

1) Detecting upper bound: Detect horizontal noisy black borders in the upper side of the image. Calculate Y_{START} as follow:

$$Y_{START} = \left\{ y_i \mid y \in \left[0, \frac{I_y}{6}\right], \max\left(H_h(y)\right) \right\} \qquad (3)$$

I_y is number of rows in the document image.

2) Detecting lower bound: Horizontal noisy black borders Y_{END} in the lower side of the image is calculated in the same way.

$$Y_{END} = \left\{ y_i \mid y \in \left[\frac{5I_y}{6}, I_y\right], \max\left(H_h(y)\right) \right\} \qquad (4)$$

D. Vertical histogram

Calculate vertical histogram H_v, which is the sum of black pixels in each column within Y_{START} and Y_{END} limits.

$$H_v(x) = \sum_{y=Y_{START}}^{Y_{END}} P(x,y) \ \ , where 1 \le x \le I_x \tag{5}$$

1) Detecting left bound: Due to existence of the edge of scanner cover in the left side of the image, calculation of this bound is different from upper and lower bounds. You can see the edge of scanner in Fig. 4.

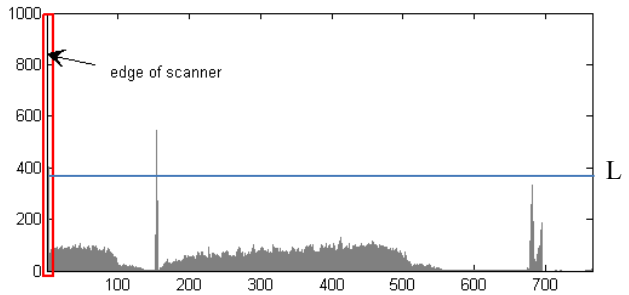


Fig. 4. Vertical histogram of Fig. 1-a

To find the edge of scanner and Left bound, we used the following equation:

$$\begin{aligned} ES &= x_i : \max\big(H_v(x)\big)\ x \in \left[0, \frac{2I_x}{3}\right] \\ &, if\big(H_{vmax} > L \ \&\ 1 < x_i < \frac{I_x}{20}\big) \\ X_{LEFT} &= x_i : \max\big(H_v(x)\big)\ x \in \left[ES, \frac{2I_x}{3}\right] \\ &, if(x_i > ES + th) \\ X_{LEFT} &= x_i : \max\big(H_v(x)\big) \quad , if(H_{vmax} < L) \end{aligned} \tag{6}$$

ES is the edge of scanner. The first condition, is satisfied when the edge of scanner detected. After that we find left limits with second condition. The third condition, is satisfied when edge of scanner is not detected or small value of this exist.

2) Detecting right bound: Vertical noisy black borders X_{RIGHT} in the right side of the image is calculated similarly to the second condition of the left bound as follows:

$$X_{RIGHT} = \left\{x_i \mid x \in \left[\frac{I_x}{3}, I_x\right], \max\big(H_v(x)\big)\right\} \tag{7}$$

E. Removing noisy area – phase 1

In the first phase, all pixels located outside of the detected bounds will be removed; we call the result as I_i:

$$\begin{aligned} \mathrm{I}_i(\mathrm{x},\mathrm{y}) &= 0, if(\mathrm{y} < Y_{START}\ or\ y > Y_{END}) \\ &\quad \&(x > X_{RIGHT}\ or\ x < X_{LEFT}) \\ \mathrm{I}_i(\mathrm{x},\mathrm{y}) &= \mathrm{I}(\mathrm{x},\mathrm{y}) \qquad , o.w \end{aligned} \tag{8}$$

Fig. 5 shows gradient map and vertical and horizontal histogram which are projected into *X* and *Y* axes. As you see there is a peak near start and end of page as shown in green rectangle.

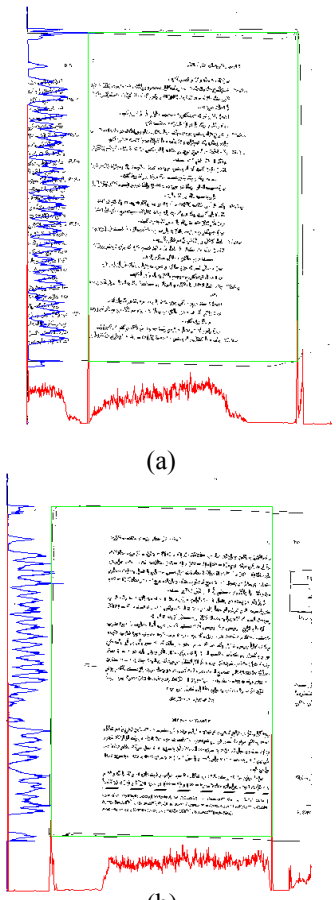
F. Removing noisy area – phase 2

1) Median of Limited Histogram: To find which side of the detected borders need to be recalculated, we find median of limited histogram. This value is calculated as follows:

$$m: \sum_{y=1}^{m} H_n(x) > 0.5 \tag{9}$$

Where

$$H_n = \left(\frac{1}{\sum_{x=x_{LEFT}}^{x_{RIGHT}} H_v(x)}\right) . H_{v_{LIMITED}} \tag{10}$$



(a)

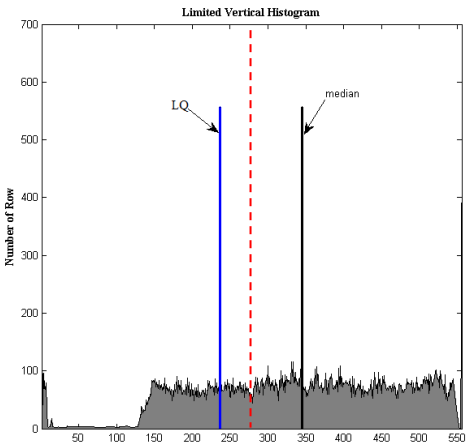
(b)

Fig. 5. Histogram of edge map. Green rectangle indicates the possible document area in phase 1. (a) Result of Fig. 1-a (b) Result of Fig. 1-b.

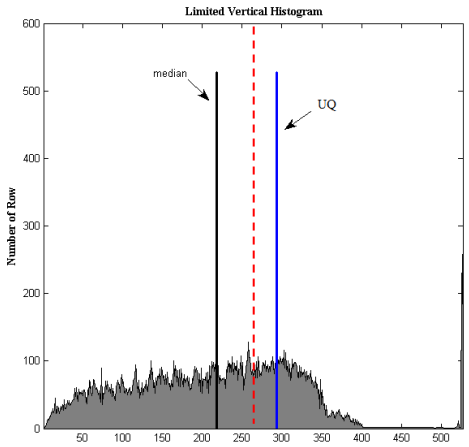
2) Detect Quartile: According to Fig. 6-a, when the noisy text regions is located in the right side of the document, median is greater than half size of the limited histogram H_n, Then left limit is calculated again to find the exact bound. For detecting fence of end data need to be examined; we use lower quartile LQ (25th percentile). Also according to Fig. 6-b, when the noisy text regions located in the left side of document, right limit must be calculated again and we use upper quartile UQ (75th percentile) for fence of starting data.

According to the following formula lower quartile LQ or upper quartile UQ calculated:

$$UQ: \sum_{UQ}^{X_{RIGHT}} H_n(x) > 0.25,$$
$$, if\ (m > \frac{X_{RIGHT} - X_{LEFT} + 1}{2}) \qquad (11)$$
$$LQ: \sum_{x=X_{LEFT}}^{LQ} H_n(x) > 0.25 \qquad o.w$$



(a)



(b)

Fig. 6. Limited vertical histogram. (a) Result of Fig. 1-a (b)Result of Fig. 1-b

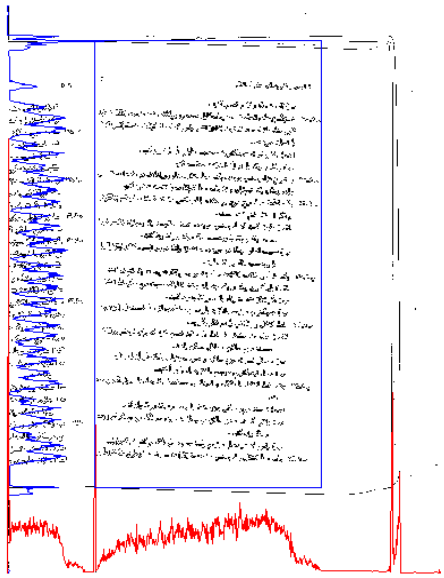
According to first condition of equation (11), if median is greater than half size of the limited histogram H_n, for calculating exact location of X_{LEFT}, first limited histogram $H_{v_{LIMITED}}$ is multiplied by rectangular windows , $X_{LEFTnew}$ is equal to the end data $\langle H \rangle_{v_{end}}$ of first window of limited histogram ⟨H⟩ which satisfied first condition of equation (12), if first condition of equation (12) didn't find new limitation, $X_{LEFTnew}$ is equal to X_{LEFT}. If median is smaller than half size of the limited histogram H_n , exact location of X_{Right} is calculated similarly.

$$X_{LEFTnew} = \langle H \rangle_{v_{end}} \left(\langle H \rangle_{v_{end}} - \langle H \rangle_{v_{first}} > TH1 \right) \\ \&(\langle H \rangle_{v_{end}} > LQ) \qquad (12)$$
$$X_{LEFTnew} = X_{LEFT} \qquad o.w$$

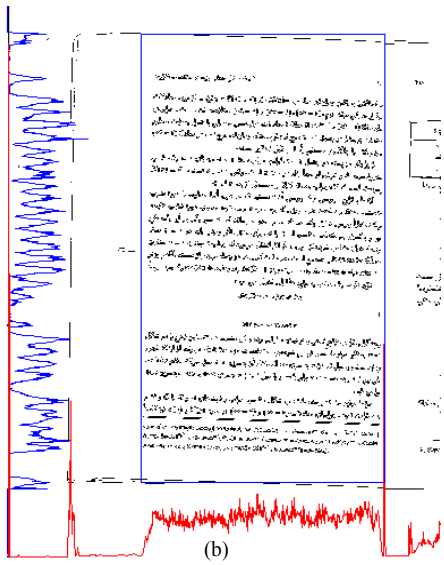
Fig. 7 shows result of Final removal noisy area. As you see blue rectangle shows that we find final bounds more exactly.

IV. EXPERIMENTAL & RESULT

To verify the validity of the proposed method, we applied the proposed method on our dataset. We used 159 document images form Persian books that contain noisy black borders as well as noisy text region appeared from adjacent pages. After visual checking, we found that in 150 images (94.33% of our samples) the noisy black borders and noisy text region were correctly removed. For 108 images (67.92% of dataset), new bounds found according to section 3-F , and in other 42pages

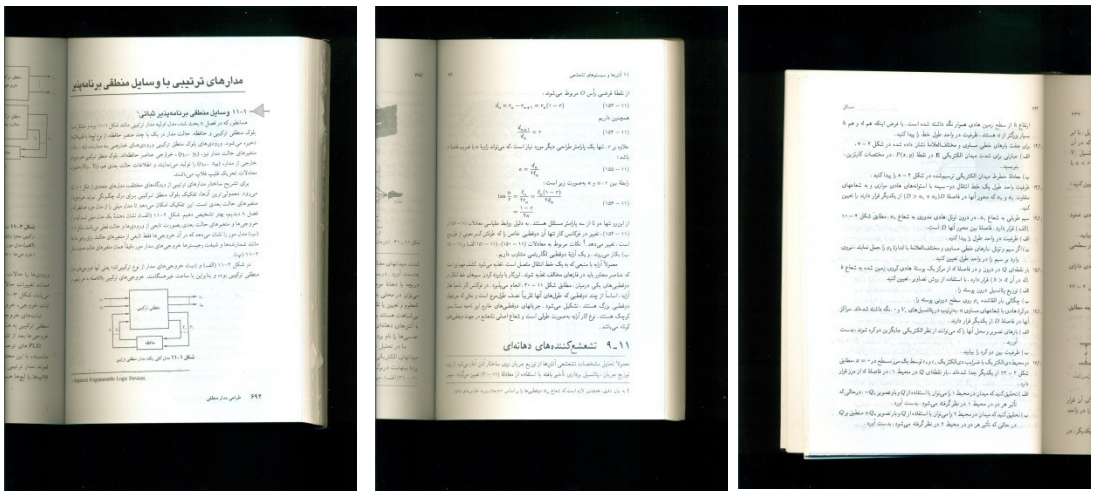


(a)



(b)

Fig. 7. Histogram of edge map. Blue rectangle indicates the possible document area in phase 2. (a) Result of Fig. 1-a (b)Result of Fig. 1-b.



(a)

(b)

Fig. 6. (a) Original camera document images, (b) result of proposed method

(26.41% of dataset) final border is equal to the result of section 3-E. Only for 9 images (5.67% of dataset) the proposed method produced bad results. Fig. 8 depicts some example of document image illustrating the page borders detection and removal process.

V. CONCLUSION

This paper proposed an effective algorihtm for detection and removal of noisy black borders and extra text regions inserted on document images scanned from large books. The new algorithm is based on projection profile combined with an edge detection process. In the future work we will focus on other kind of documents such as newspaper, two-page documents, camera captured documents taken with mobile phones and optimization of the proposed method.

REFRENCES

[1] L. A. Fletcher and R. Kasturi, “A robust algorithm for text string separation from mixed text/graphics images,” IEEE Trans. on PAMI Vol. 10, pp. 910-918, Nov 1988.

[2] K. C.Fan,Y.K Wang, and T.R Lay, “Marginal Noise Removal of Document Images”, Pattern Recognition, vol. 35, pp. 2593-2611, 2002.

[3] B. T. Avila, and R.D. Lins, “A New Algorithm for Removing Noisy Borders from Monochromatic Document”, Proc. of ACM-SAC’2004, Cyprus, ACM Press, March 2004

[4] B. T. Avila, and R.D. Lins, “Efficient Removal of Noisy Borders from Monochromatic Documents”, ICIAR 2004, Vol. 3212, pp. 249-256, 2004.

[5] D.X. Le,and G.R. Thoma, “Automated Borders Detectionand Adaptive Segmentation for Binary Document Images”,International Conference on Pattern Recognition, vol. 3, pp. 737-741, 1996.

[6] N. Stamatopoulos, B. Gatos, andA. Kesidis, “Automatic Borders Detection of Camera Document Images”, 2nd International Workshop on Camera-Based Document Analysis and Recognition, Curitiba, Brazil,pp. 71-78, 2007.

[7] F.Shafait, J. Beusekom, D. Keysers, andT.M. Breuel, “Document cleanup using page frame detection”, Int. Jour. OnDocument Analysis and Recognition, vol. 11, pp. 81–96, 2008.

[8] A.Jain, Fundamentals of Digital Image Processing, Prentice-Hall, Englewood Cliffs, NJ, 1989.

[9] R.C.Gonzalez, R.E. Woods, Digital Image Processing, Addison Wesley, 2002